GGG-S-62D
October 24, 1986
(SUPERSEDING)
GGG-S-62C
June 25, 1981

FEDERAL SPECIFICATION

SAW, CHAIN, PORTABLE, POWER

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers pneumatic-motor-driven, gasoline-engine-driven and electric-motor-driven portable chain saw.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the titles of the specifications but include only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. Chain saws shall be of the following types, and classes, as specified (see 6.2):

Type I - Pneumatic-motor-driven.
Type II - Gasoline-engine-driven.
    Class A - General purpose.
    Class B - Lightweight.
    Class C - Heavy duty.
Type III - Electric-motor-driven.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-C-320	- Chromium Plating (Electrodeposited).
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberwood.

AMSC N/A FSC 3695
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Federal Standards:

FED-STD-101	- Test Procedures for Packaging Material.
FED-STD-123	- Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Description from established distribution points in their agencies.)

Military Specifications:

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-W-6858	- Welding Resistance, Spot and Seam.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-W-8611	- Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys, Process For.
MIL-E-10062	- Engines; Preparation for Shipment and Storage of.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof Flexible.
MIL-C-12044	- Chests, Plywood.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1474	- Noise Limits for Army Materiel.

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(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR

Federal Regulations:

29 CFR Chapter XVII, Part 1910 and Amendments - Occupational Safety and Health Administration Standards.

(The Code of Federal Regulations [CFR] and the Federal Register [FR] are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.
E 18 - Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC (AWS)

A5.13 - Surfacing Welding Rods and Electrodes.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 NW 7th Street, Miami, FL 33125.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

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(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI B175.1 - Gasoline Powered Chain Saws Safety Requirements.

(Application for copies should be addressed to the American National Standard Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. Each chain saw shall consist essentially of a lightweight power unit, controls, power transmission housing, cutter guide bar, cutting chain, chain-adjusting device, oiling device, and operating handles.

3.2 First article. Unless otherwise specified (see 6.2) a sample shall be subjected to first article inspection (see 4.3 and 6.6).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Recycled material may be used provided it meets all of the requirements of this specification (see 6.5).

3.3.1 Material deterioration prevention and control. The chain saw shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the chain saw may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4 Safety. Guards, switches, controls and other appropriate protection shall be provided for the operator in accordance with all applicable Federal Occupational Safety and Health Regulations.

3.4.1 Chain brake and hand guard. Type I chain saws shall be equipped with a hand guard meeting the requirements specified in ANSI B175.1. Type II and III chain saws shall be equipped with a chain brake meeting the requirements specified in ANSI B175.1.

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3.5 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement.

3.6 Power-transmission housing. The power-transmission housing shall contain the clutch, driving sprocket, and shall form the cutter guide bar head.

3.6.1 Bumper spike. The bumper spike, shall be of toothed construction to afford a nonslip grip against the log and shall extend below the lower edge of the cutting chain.

3.6.2 Chain drive sprocket. The chain drive sprocket shall be of an alloy steel and shall be heat treated to a hardness of not less than 54 Rockwell C, and shall withstand the performance testing as specified herein. Sprocket pitch shall match the pitch of the saw chain. The sprocket diameter shall be matched to the guide bar width to provide a smooth chain entry.

3.7 Guide bar. The guide bar shall be of high carbon steel alloy and shall be grooved and contoured to insure smooth chain travel. The groove shall be centered throughout the bar length, funneled to permit smooth chain entry, and shall be of the correct width and depth for the cutter chain furnished. The bar rails shall be heat treated to a hardness of not less than 55 Rockwell C. The remainder of the bar shall be not less than 35 Rockwell C. When laminated bars are furnished, the laminations shall be secured by plug or spot welds, spaced not less than 1 inch on center around the bar rails and nose section and not greater than 1-1/2 inches in the remainder of the bar or by copper braze. Plug and spot welding shall be in accordance with MIL-W-8611, spot resistance welding in accordance with MIL-W-6858, and brazing in accordance with MIL-B-7883. Unless otherwise specified, sprocket nose guide bars shall be furnished. The sprocket wheels shall be supported by antifriction bearings. The sprocket shall be thin enough to pass through the cut and shall be replaceable without the use of special tools. A lubricating gun, when required, shall be supplied with the saw to lubricate the sprocket. When specified (see 6.2), bow guide bars or solid nose bars shall be furnished. The nose shall have a built-up hard surfacing of 0.04 inch minimum thickness. The hard surfacing rod shall conform to AWS A 5.13; AWS Classification RCoCr-A, or RCoCr-B, ECoCr-A.

3.8 Cutter chain. The chain saw shall be equipped with a cutter chain that will permit sharpening with a straight round file without removal from the guide bar. The cutter teeth, tie bars and drive tangs shall be heat-treated to a hardness of not less than 49 Rockwell C. Unless otherwise specified herein, the cutter teeth shall be plated with a hard chromium plating conforming to QQ-C-320, class 2. The chain pitch shall be the option of the contractor and shall be subject to the performance requirements specified herein. Chain shall be of the new technology low kickback design.

3.9 Chain-adjusting device. A chain-adjusting device shall be provided to positively extend the guide bar to maintain correct chain tension. The lug shall positively engage the guide bar and shall not become disengaged under any condition of operation as specified herein. The adjusting device shall have sufficient travel to permit removal of the cutter chain from the guide bar without disassembly of the chain or removal of the guide bar.

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3.10 Handles. The chain saw shall be equipped with handles that meet the requirements of ANSI B175.1.

3.10.1 Handle vibration isolation system. Type I and II saws shall be equipped with anti-vibration handles. The shock absorbers in the anti-vibration handles shall be replaceable with disposable shock absorbers. The handles shall meet the vibration requirements of ANSI B175.1.

3.11 Lubrication.

3.11.1 Chain oiling system. Type I and II chain saws shall be provided with automatic chain oiling system only; type III chain saws shall have automatic or manual oiling systems. The manual-operated system shall be thumb-operated without requiring removal of the operator's hands from the handles or throttle control.

3.11.1.1 Reservoir. The chain oil reservoir shall have sufficient capacity for not less than 15 minutes of sawing operations. The filler-opening cap shall be a spillproof type. The cap, or the reservoir immediately adjacent to the cap, shall be plainly marked to indicate chain oil; the cap shall be secured against loss.

3.11.2 Enclosures with integral reservoir of lubricant. Each enclosure shall be equipped with a means for filling the enclosure with lubricant. Each enclosure, when filled with lubricant, shall not permit loss of lubricant with the chain saws in all positions. Accessibility to the filler shall be obtained without the removal or adjustment of accessories or parts.

3.12 Noise limits. The noise produced by the chain saw shall conform with MIL-STD-1474 requirements when tested in accordance with specification, 4.5.2.1.9. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3 and be readable at the distance indicated on the sign.

3.13 Identification marking. The saw shall be legibly and durably marked in accordance with MIL-STD-130.

3.14 Treatment and painting. The sections of the chain saws and their components normally painted shall be cleaned, treated, and painted in accordance with the contractor's standard practice.

3.15 Type I, pneumatic-motor-driven.

3.15.1 Pneumatic-motor-driven. The type I pneumatic-motor-driven chain saws shall operate as specified herein under water when tested as specified in 4.5.2.1.4. The transmission housing shall be fitted with watertight seals to prevent the entry of water into the housing. The oil reservoir shall be fitted with an unvented cap to prevent the entry of water into the reservoir.

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3.15.2 Air motor. The air motor shall be of the rotary (vane) positive displacement type and shall develop not less than 3.5 horsepower with an air consumption of not more than 90 cfm at 90 psi. The air motor shall withstand 100 hours of full-load operation without loss of efficiency. The exhaust of the motor shall be directed so as to blow away from the operator. A spring-loaded, (closed position), air-control valve shall be incorporated in the operating handle. The air inlet and exhaust shall have 3/4 inch, internal-taper, pipe threads for attaching 3/4 inch pneumatic-hose fittings. The air-exhaust port and inlet port shall be permanently identified.

3.15.3 Size. The size shall be the effective cutting capacity as defined by figure 1 and 2 as applicable. Size shall be not less than 18 inches or more than 40 inches as specified (see 6.2).

3.15.4 Weight. The empty weight of the chain saw shall be not more than 19 pounds. Empty weight is the weight of the power unit complete but not including the weight of the guide bar, chain, spikes, fuel, chain lubricating oil or accessories which are optional or sold separately.

3.15.5 Cutter chain. The cutter chain shall conform to the requirements specified in 3.8.

3.16 Type II gasoline-engine-driven.

3.16.1 Power unit. The power unit shall be a lightweight, air-cooled, 2-cycle gasoline engine. The engine shall withstand 75 hours of full load or cyclic load operation and shall develop sufficient power to meet all performance requirements specified herein for the applicable class saw, with not more than two spark plug changes or cleanings and two cylinder port cleanings.

3.16.1.1 Bearings. The engine crankshaft shall be supported by ball or roller antifriction bearings at each end of the crankshaft. Needle bearings shall be utilzed at both the crankpin and piston pin ends of the connecting rod, except sleeve bearings are permitted at the piston pin on type 2, class B saws.

3.16.1.2 Carburetor. The carburetor shall enable the engine to operate in all positions. The carburetor shall be equipped with a fuel strainer at the fuel inlet and shall have an adjustable idle jet for adjusting the air-fuel ratio at idle speed.

3.16.1.3 Air cleaner. The carburetor air inlet shall be fitted with an air cleaner to prevent the entrance of dirt and dust into the engine. The element shall be accessible and removable without use of special tools and shall not be adversely affected by cleaning in solvents.

3.16.1.4 Ignition system. The engine ignition system shall be solid state and shall develop sufficient voltage to enable the engine to start and operate under all conditions specified herein. Ignition wiring shall be secured to prevent chafing from vibration or moving parts.

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3.16.2 Controls. The chain saw shall be equipped with a throttle control, choke control or engine primer, and an ignition switch. The throttle control shall be located in the rear handle, and operable without the operator removing his hands from the saw handles, engine RPM shall return to idle when throttle in released. The choke control shall be clearly and permanently marked to indicate the choke position. The ignition switch shall be so located that it can be moved to the "OFF" position by the operator, without removal of the hands from the handles. The "OFF" position shall be clearly and permanently marked.

3.16.3 Fuel tank. The fuel tank shall be nonferrous metal, treated on the inside to prevent corrosion, nylon or high density polyethylene. The tank shall have sufficient capacity to permit not less than 15 minutes continuous operation at maximum power. Construction and mounting of the fueltank shall be such that neither vibration nor excessive tightening of the mounting provisions (when used), will cause leaks. The fuel tank shall have a leakproof vent to permit operation of the chain saw in all positions without leakage, except when the saw is in the inverted position, some seepage is acceptable. The fill cap shall be secured against loss.

3.16.4 Exhaust system. The exhaust system shall be equipped with a spark arressting muffler to reduce the exhaust noise and direct the exhaust gases away from the operator. The muffler shall be so located as not to constitute a hazard to the operator, exhaust system shall meet the requirements of ANSI B175.1.

3.16.5 Clutch. The clutch shall be of the centrifugal, self-energizing friction type. The clutch shall transmit sufficient torque to enable the saw to meet all performance requirements specified herein.

3.16.6 Cranking system. Cranking shall be accomplished manually by means of an automatic rewind starter. The starter shall incorporate a positive means of engagement. Friction pawls, shoes, or wedges shall not be permitted. The starter shall withstand, without failure, slippage or starter rope breakage, all engine starts required when the saw is tested as specified herein.

3.16.7 Speed limitation. A governor shall be furnished if required to prevent damage by momentary maximum speed at no load.

3.16.8 Fuel can. When specified (see 6.2), a 2-1/2-gallon fuel can shall be provided. The fuel can shall be equipped with a self-storing, flexible pour spout with strainer. The spout diameter shall be small enough to fit inside the fuel tank fill opening. The fill cap shall incorporate a graduated container to enable accurate measurement of oil for the manufacturer's recommended gasoline-oil mixture.

3.16.9 Lubrication. The power unit shall be lubricated by means of a gasoline-oil fuel mixture. The manufacturer's recommended gasoline-oil mixture oil shall be plainly indicated on the chain saw.

3.16.10 Type II-gasoline-engine-driven. Chain saws shall have the cutter chain sprocket directly driven from the engine crankshaft, through a centrifugal clutch.

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3.16.10.1 Class A, general purpose.

3.16.10.1.1 Weight. The empty weight of the chain saw shall be not more than 15 pounds. Empty weight is the weight of the power unit complete, but not including the weight of the guide bar, chain, spikes, fuel, chain lubricating oil or accessories which are optional or sold separately.

3.16.10.1.2 Size. The size shall be the effective cutting capacity as defined by figure 1 and 2 as applicable. Size shall be not less than 16 inches or more than 24 inches as specified (see 6.2).

3.16.10.1.3 Performance. The cutter chain shall conform to the requirement specified in 3.8. The saw shall establish a minimum sawing rate of not less than 300 sq. ft./hr. in oak logs and shall maintain not less than 50 percent of the maximum rate established through an accumulated area of 450 sq. ft. without resharpening.

3.16.10.1.4 Engine size. Engine displacement shall be not less than 3.5 cubic inches.

3.16.10.2 Class B, lightweight.

3.16.10.2.1 Weight. The empty weight of the chain saw shall be not more than 10.5 pounds. Empty weight is the weight of the power unit complete, but not including the weight of the guide bar, chain, spikes, fuel, chain lubricating oil or accessories which are optional or sold separately.

3.16.10.2.2 Size. The size shall be the effective cutting capacity as defined by figure 1 and 2 as applicable. Size shall be not less than 10 inches or more than 16 inches as specified (see 6.2).

3.16.10.2.3 Performance. The cutter chain shall conform to the requirements specified in 3.8. The saw shall establish a minimum sawing rate of not less than 200 sq. ft./hr. in oak logs and shall maintain not less than 50 percent of the maximum rate established through an accumulated area of 300 sq. ft. without resharpening.

3.16.10.2.4 Engine size. Engine displacement shall be not less than 2.3 cubic inches.

3.16.10.3 Class C, heavy duty.

3.16.10.3.1 Weight. The empty weight of the chain saw shall be not more than 23 pounds. Empty weight is the weight of the power unit complete, but not including the weight of the guide bar, chain, spikes, fuel, chain lubricating oil or accessories which are optional or sold separately.

3.16.10.3.2 Size. The size shall be the effective cutting capacity as defined by figure 1 and 2 as applicable. Size shall be not less than 22 inches or more than 36 inches as specified (see 6.2).

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3.16.10.3.3 Performance. The cutter chain shall conform to the requirements specified in 3.8. The saw shall establish a minimum sawing rate of not less then 450 sq. ft./hr. in oak logs and shall maintain not less than 50 percent of the maximum rate established through an accumulated area of 550 sq. ft.

3.16.10.3.4 Engine size. Engine displacement shall not be less than 5.0 cubic inches.

3.17 Type III, electric-motor-driven.

3.17.1 Electric motor. The electric motor shall have a minimum of 1.5 horsepower, operate on 115 volt, 60 hertz ac power, and shall develop sufficient power to meet the performance requirements specified herein. Motor control switch shall be spring loaded to off position.

3.17.2 Weight. The empty weight of the chain saw shall be not more than 12 pounds. Empty weight is the weight of the power unit complete, but not including the weight of the guide bar, chain, spikes, fuel, chain lubricating oil or accessories which are optional or sold separately.

3.17.3 Size. The size shall be the effective cutting capacity as defined by figures 1 and 2 as applicable. Size shall be not less than 8 inches or more than 14 inches, as specified (see 6.2).

3.17.4 Performance. The cutter chain shall conform to the requirements specified in 3.8. The saw shall establish a minimum sawing rate of not less than 100 sq. ft./hr. in oak logs and shall maintain not less than 50 percent of the maximum rate established through an accumulated area of 250 sq. ft. without resharpening.

3.18 Military requirements.

3.18.1 Accessory kit. When specified (see 6.2) each chain saw shall be furnished with the accessory kit designated in level I or level II, consisting of the following items:

LEVEL I

QUANTITY	DESCRIPTION
1 ea	Combination screwdriver, spark plug and bar nut wrench.
2 ea	Chain saw files of the proper diameter for the chain pitch furnished.
1 ea	File Guide.
1 ea	Spark Plug (Type II only)
1 ea	Scabbard
1 ea	Chain of the type and sign furnished with the chain saw.

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LEVEL II Includes items in level I plus the following:

1 ea	Chain repair kit.
1 ea	Pocket Chain breaker.
1 ea	Plastic wedge, buckling approximately 5-1/4 inches long.
1 ea	Plactic wedge, falling approximately 8 inches long.
2 qts	Bar and chain oil.
2 qts	Two-cycle oil.

3.18.2 Chest. When specified (see 6.2), the chain saw with all tools, and gasoline can, and repair parts shall be enclosed in a chest conforming to MIL-C-12044. The chest shall hold the contents in place during transit and shall withstand a free full drop when tested as specified in 4.5.2.1.11. Chain saws equipped with quick disconnects between the power unit and the saw attachment shall be placed in the chest in a knockdown position.

3.19 Operating and safety instructions. Each chain saw shall be equipped with operating and safety instructions as specified (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article shall be subjected to the tests marked "X" of table I. Failure of any test shall be cause for rejection.

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TABLE I. First article tests.

Type I	Type II	Type III	Test	Requirement
X	-	-	4.5.2.1.2	3.15.2
X	-	-	4.5.2.1.3	3.15.2
X	-	-	4.5.2.1.4	3.15.1
X	X	X	4.5.2.1.5	3.4.1
X	X	X	4.5.2.1.5.1	3.4.2
-	X	X	4.5.2.1.6	3.16.10.1.3
-	X	X	4.5.2.1.7	3.16.10.2.3, 3.16.10.3.3,
-	X	-	4.5.2.1.8	3.16.1
X	X	-	4.5.2.1.9	3.12
X	X	X	4.5.2.1.10	3.6, 3.7, 3.8, 3.9, 3.11, 3.15, 3.16, and 3.17
X	X	-	4.5.2.1.11	3.18.2
X	X	X	4.5.2.1.12	3.6.2, 3.7, 3.8
-	X	-	4.5.2.1.13	3.16.4
X	X	X	4.5.2.1.14	3.10
X	X	-	4.5.2.1.15	3.10.1

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. The lot shall consist of all saws of one type and class offered for delivery at one time.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Type I.

4.4.3.1.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1.10 and 4.5.2.1.2. AQL shall be 1.0 percent defective. Inspection level shall be S-2.

4.4.3.2 Type II.

4.4.3.2.1 Samples. Samples selected in accordance with 4.4.1, shall be tested as specified in 4.5.2.1.10 and 4.5.2.1.6. AQL shall be 1.0 percent defective. Inspection level shall be S-2.

4.4.3.3 Type III.

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4.4.3.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1.10 and 4.5.2.1.6. AQL shall be 1.0 percent defective. Inspection level shall be S-2.

4.5 Inspection procedure.

4.5.1 Examination. Examine the chain saw for the following defects:

Major

101. Weight not as specified.
102. Assembly not complete or not as specified.
103. Parts or components missing.
104. Safety devices or guards missing or not as specified.
105. Bumper spike not as specified.
106. Drive sprocket not as specified.
107. Guide bar not as specified.
108. Cutting capacity not as specified.
109. Chain-adjusting device not as specified.
110. Chain oiling system not as specified.
111. Lubricant reservoir not as specified.
112. Air motor not as specified (type I only).
113. Gasoline engine and components not as specified (type II only).
114. Electric motor not as specified (type III only).
115. Cutter chain not as specified.
116. Plating on cutter teeth not as specified.
117. Material not as specified.
118. Materials not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration.
119. Dissimilar metals not in accordance with MIL-STD-889.
120. Material and material finish or treatment information not available.
121. Operating or safety instruction missing.
122. Accessory kit not as specified.

Minor

201. Treatment and painting not as specified.
202. Identification marking not as specified.
203. Chest not as specified.

4.5.2 Tests.

4.5.2.1 Test material. Oak logs used as test material shall be green having a density of 64-71 pounds per cubic foot.

4.5.2.1.1 Run-in. Prior to conducting any test, chain saws shall be run in, checked and serviced in accordance with the contractor's standard procedure. The cutter chains shall be run-in, resharpened and adjusted as required.

4.5.2.1.2 Brake horsepower. Test the air motor with a dynamometer to determine the bhp. Nonconformance to 3.15.2 shall constitute failure of this test.

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4.5.2.1.3 Endurance. Test the chain saw for 100 hours at continuous load operation at minimum horsepower specified. The integral oiling device of the air motor shall be the only source of lubrication. Failure of the chain saw to operate for 100 hours as specified or damage to or permanent deformation of components or parts thereof shall constitute failure of this test.

4.5.2.1.4 Submersion. Operate the chain saw making 10 cuts in succession in an 18-inch-diameter oak log submerged in water. Six of the cuts shall be crosscuts through the log and four shall be rip cuts for a distance of not less than 3 feet. The saw shall be completely submerged in the water during this test. Failure of the saw to complete the test, malfunctioning, or damage to or permanent deformation of components, shall constitute failure of this test.

4.5.2.1.5 Chain brake and hand guard test. Hand guard and chain brake shall be tested in accordance with the procedures outlined in ANSI B175.1. Nonconformance to 3.4.1 shall constitute failure of this test.

4.5.2.1.5.1 Kickback test. Test the chain saw in accordance with the kickback requirements in ANSI B175.1. Nonconformance to 3.4.2 shall constitute failure of this test.

4.5.2.1.6 Maximum production. Operate the saw making five complete cutoffs in a select oak log, which will be used as a control log, timing each cut with a stop watch. The average area of the five cuts divided by the average time of the five cuts shall establish the maximum production rate. A rate less than that specified for the applicable type and class saw shall constitute failure of this test. The average engine speed at which this rate was established shall be recorded.

4.5.2.1.7 Chain dulling. Continue to operate the chain saw, cutting logs from a log pile without resharpening the chain. Determine the production rate, in accordance with the procedure in 4.5.2.1.6 at intervals of 50 sq. ft. of accumulated area, including the area of the timed cuts. Repeat this procedure until a total accumulated area of 300 to 550 sq. ft. as applicable to type and class saw has been sawed. A final production rate at the completion of the applicable square footage of area of sawing of less than 50 percent of that established in 4.5.2.1.6, shall constitute failure of this test.

4.5.2.1.8 Endurance. The endurance test shall be conducted at constant speed, full load or on a cylic schedule at the option of the contractor.

4.5.2.1.8.1 Constant speed full load. Test the power unit on a suitable load absorption device for 75 hours at constant load with wide-open throttle. Adjust the load to maintain the engine speed to that established in 4.5.2.1.6. Inability of the power unit to operate for 75 hours as specified, or damage to or permanent deformation of parts or components shall constitute failure of this test.

4.5.2.1.8.2 Cyclic load. Test the power unit on a suitable load absorption device equipped with the necessary controls to alter the speed and load in the following sequence:

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(1) Idle speed - no load.
(2) Momentary wide open throttle - no load.
(3) Wide open throttle - full load.
(4) Momentary wide open throttle - no load.

Repeat this cycle until 75 hours have been accumulated. The time intervals for each segment shall be selected by the contractor, but shall be proportioned to simulate actual bucking operations.

4.5.2.1.9 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load.

4.5.2.1.10 Operation test. Operate the chain saw for not less than 15 minutes to determine the completness and correctness of assembly. Inability to operate or any malfunction, breakage, or permanent deformation of any part or component; or failure of any control to operate as required shall constitute failure of this test.

4.5.2.1.11 Chest test. Subject the chest equipped as specified in 3.18.2 to the free-fall drop test in accordance with test method 5007 of FED-STD-101. damage to the components within the chest, or any loosening, breakage, displacement to any blocking, bracing, or anchoring shall constitute failure of this test.

4.5.2.1.12 Hardness test. Rockwell hardness test shall be conducted for the requirements of 3.6.2, 3.7, and 3.8 in accordance with ASTM E18. Failure to meet these requirements shall constitute failure of this test.

4.5.2.1.13 Exhaust system test. Test the exhaust system in accordance with ANSI B175.1. Nonconformance to 3.16.4 shall constitute failure of this test.

4.5.2.1.14 Operating handles. Handles shall be tested as specified in ANSI B175.1. Nonconformance to 3.10 shall constitute failure of this test.

4.5.2.1.15 Handle vibration isolation system. Saw shall be tested for vibration as specified in ANSI B175.1. Nonconformance to 3.10.1 shall constitute failure of this test.

4.6 Inspection of packaging. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 1.0 percent defective for major defect and 4.0 percent defective for minor defects.

4.6.1 Examination for military levels of protection. Samples selected in accordance with 4.6 shall be examined for the following defects.

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Major

123. Materials, methods, or containers not as specified. Each incorrect material, method, or container shall be considered one defect.
124. Disassembly not as specified.
125. Oil reservoir and fuel can not coated with preservative as specified.
126. Cutter guide bar not coated with preservative and wrapped, and the wrap not secured as specified.
127. Cutting chains not preserved as specified.
128. Air motors not preserved and air inlet and exhaust port not closed as specified.
129. Gasoline engine not preserved as specified.
130. Consolidation not as specified.
131. Case liner not provided as specified.
132. Chest lid not secured as specified.
133. Marking illegible, incorrect, incomplete or missing.

Minor

204. Repair parts and maintenance tools not preserved and packaged in accordance with MIL-P-116 as specified.
205. Technical publications not preserved in accordance with MIL-P-116, method IC-1.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be level A, or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The cutter guide bar and cutting chain shall be removed.

5.1.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in, and shall be applied in accordance with MIL-P-116.

5.1.1.3 Chain oiling system. The interior surfaces of the reservoir and threads of the fill cap shall be coated with type P-10, preservative (grade 30) and the excess preservative drained.

5.1.1.4 Fuel can. The fuel can shall be preserved as specified in 5.1.1.4 for the chain oiling system.

5.1.1.5 Cutter guide bar. The cutter guide bar shall be coated with type P-10 (grade 30) preservative. The bar shall then be wrapped with barrier material conforming to MIL-B-121, type II, grade A, class 1 or 2. Class 1 materials shall be heat sealed. Class 2 material shall be sealed.

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5.1.1.6 Cutting chains. The cutting chains shall be preserved in accordance with MIL-P-116, method IC-1, and sufficient amount of type P-3 or P-9 preservative shall be applied to assure penetration of the preservative to the inner surfaces of the pins. The excess preservative shall be drained prior to enclosing in the bag.

5.1.1.7 Pneumatic motors. Interior surfaces of pneumatic motors shall be coated with type P-10, (grade 30) preservative, by feeding the preservative into the air inlet while operating the motor, until the preservative appears at the exhaust port. The air inlet and exhaust port shall be sealed with plastic or metal plugs.

5.1.1.8 Gasoline engine. The gasoline engine shall be preserved in accordance with the level A requirements of MIL-E-10062, as specified for type II, method I.

5.1.1.9 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1.

5.1.1.10 Consolidation.

5.1.1.10.1 When chests are not procured. Each chain saw and accessories, repair parts, tools, and publications shall be consolidated in the contractor's carton or in a close-fitting box conforming to PPP-B-636, type CF, class domestic, variety SW, grade as applicable to the weight of the contents. The contents shall be blocked, braced, or cushioned as applicable to prevent movement and damage. Closure of the boxes shall be in accordance with the appendix to PPP-B-636.

5.1.1.10.2 When chests are procured. Each chain saw and accessories, repair parts, tools, and publications shall be consolidated in the chest and the chest lid closed and secured. Necessary blocking and bracing shall be used as applicable to prevent movement or damage during shipment, handling, and storage.

5.1.2 Commercial.

5.1.2.1 Military agencies. Commercial preservation shall be in accordance with ASTM D 3951.

5.1.2.2 Commerical agencies. Each complete chain saw with publications, repair parts, and accessories shall be indivdually packaged in accordance with normal commercial practice. The complete package shall be designed to protect the package items against damage during shipment, handling, and storage.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Chain saws furnished without chests. Each chain saw furnished without chest, preserved and packaged as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional. The box

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shall be provided with a case liner conforming to MIL-L-10547, type I, grade B. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.1.2 Chain saw furnished with chest. Each chain saw furnished with chest, preserved as specified in 5.1, shall be packed in close-fitting box conforming to PPP-B-636, V3c, style optional. Closure shall be in accordance with the appendix to PPP-B-636.

5.2.2 Level B.

5.2.2.1 Chain saw furnished without chest. Each chain saw furnished without chest, preserved and packaged as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-636, V3c, style optional. Closure water-proof sealing, and reinforcement of the box shall be in accordance with PPP-B-636.

5.2.2.2 Chain saw furnished with chest. Each chain saw furnished with chest, preserved as specified in 5.1, shall not require packing; but the chest lid shall be sealed to prevent accidental opening and in a manner to prevent pilferage.

5.2.3 Commercial.

5.2.3.1 Military agencies. Commercial packing shall be in accordance with ASTM D 3951.

5.2.3.2 Civil agencies. Each chain saw packaged as specified in 5.1, shall be packed in a manner to assure carrier acceptance and delivery at destination at lowest rates in containers complying with Uniform Freight Classification Rules and National Motor Freight Classification Rules.

5.3 Marking.

5.3.1 Military agencies. Marking for military levels of protection shall be in accordance with MIL-STD-129. Commercial marking shall be in accordance with ASTM D 3951 and in addition the weight and cube shall be indicated on the shipping container.

5.3.2 Civil agencies. Marking shall be in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The portable chain saws are intended for one-man operation in felling trees, bucking logs, and cutting timber.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class of saw required (see 1.2).
c. When bow guide bars are required (see 3.7).
d. When solid nose guide bars are required (see 3.7).

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e. When a first article is required for inspection and approval, and the number of units required (see 3.2).
f. Cutting capacity required (size) (see 3.15.3, 3.16.10.1.2, 3.16.10.2.2, 3.16.10.3.2, and 3.17.3).
g. When a fuel can shall be provided (see 3.16.8).
h. When the saw shall be provided with a chest (see 3.18.2).
i. Degree of preservation and degree of packing required (see 5.1 and 5.2).
j. When other than finish B strapping is required (see 5.2.1.1).
k. When accessory kit is required (see 3.18.1).

6.3 Data requirements. The contracting officer should include requirements for such data as technical manuals, instructional materials, illustrated parts lists, and contractor's maintenance and operational manual to be furnished with each chain saw.

6.4 Lubricants. The contracting officer should furnish a list of Military lubricants applicable to the chain saws covered by this specification as contained in Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.6 First article. When a first article inspection is required, the items should be a initial production sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article test results and disposition of the first article.

6.7 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

MILITARY INTERESTS:

Custodians
Army - ME
Navy - YD
Air Force - 99

Review activities
Air Force - 84
Army - MD, GL
DLA - GS

User activities
Navy - MC
Army - ER

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
USDA - AFS
INT - BLM

PREPARING ACTIVITY:
Army - ME

Project 3695-0010

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GUIDE BAR

THE EFFECTIVE CUTTING CAPACITY OF A GUIDE BAR IS DEFINED AS THE LENGTH OF CUT A GUIDE BAR AND CHAIN WILL MAKE WHEN THE GUIDE BAR ADJUSTMENT IS SET AT THE HALF-WAY POINT. THE CHAIN USED FOR THE MEASUREMENT WILL BE THE ONE NORMALLY INTENDED TO BE USED WITH THE BAR. THE MEASUREMENT WILL BE MADE FROM THE BUMPER PLATE (WITHOUT SPIKES) TO THE END OF THE CHAIN AND WILL BE EXPRESSED IN TERMS OF THE NEAREST WHOLE UNIT OF MEASUREMENT, (IN., cm).

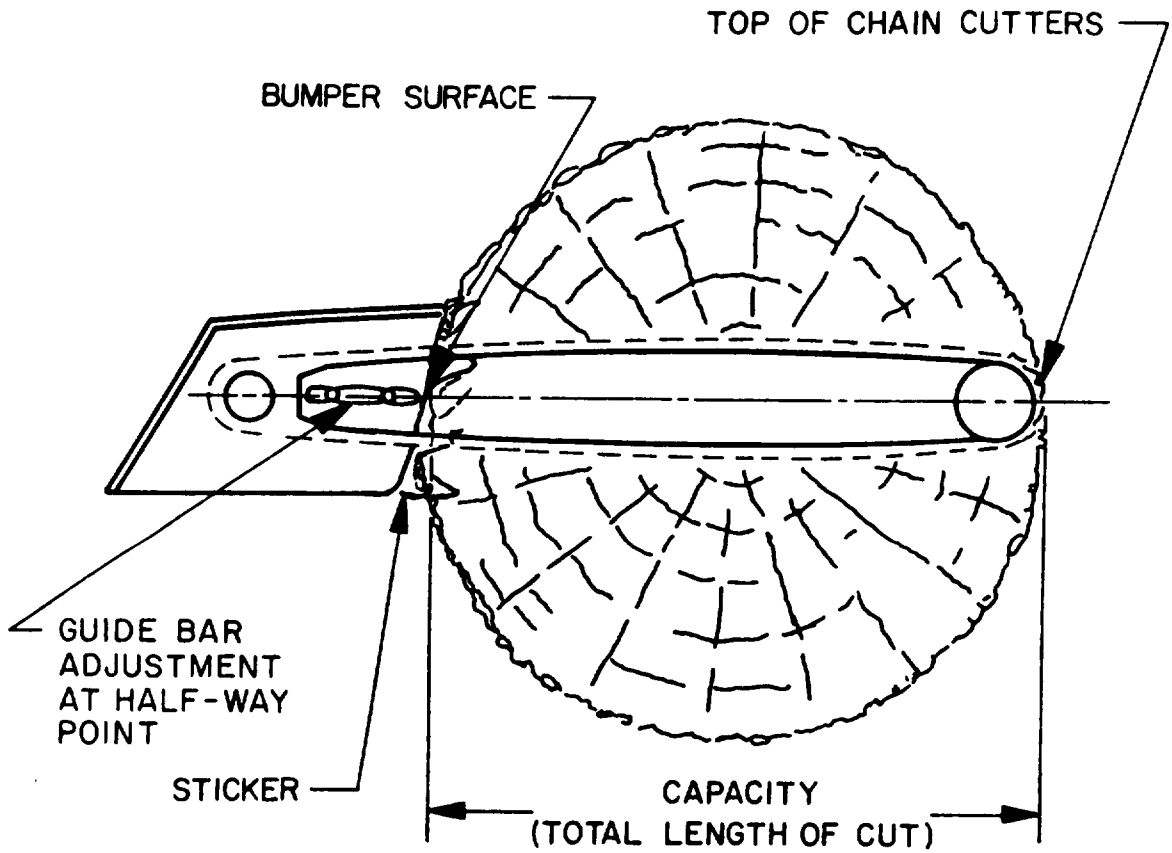


FIGURE I. GUIDE BAR

X-4529

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BOW GUIDE

THE EFFECTIVE CUTTING CAPACITY OF A BOW GUIDE IS DEFINED AS THE MAXIMUM DIAMETER LOG THAT THE BOW GUIDE,WITH CHAIN WILL CUT THROUGH. THE MEASUREMENT WILL BE MADE FROM THE FRONT SPIKES TO THE TOP OF THE CHAIN. THE MEASUREMENT WILL BE EXPRESSED IN TERMS OF THE NEAREST WHOLE UNIT OF MEASUREMENT, (IN., cm).

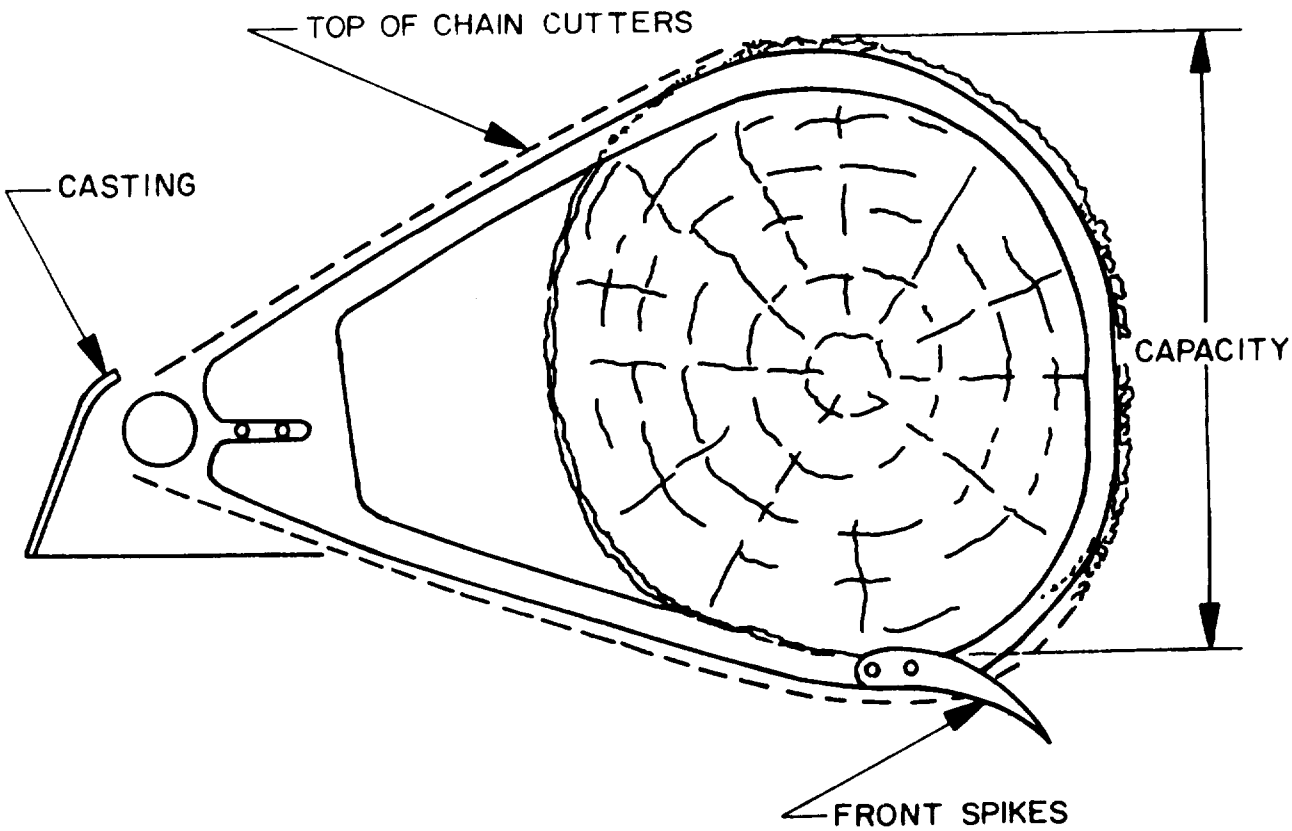


FIGURE 2. BOW GUIDE

X-4530

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GGG-S-62D	Saw, Chain, Portable, Power

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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